



Taller práctico de Git

Crear una simulación de “página web” para una cafetería llamada Café Aurora.

No usaremos código complejo.

El objetivo es aprender Git paso a paso como en un entorno profesional.

Usaremos:

- ramas
- commits
- merge
- stash
- tags
- remotos
- conflictos
- restauración
- historial
- rebase
- etc.

Caso práctico

Tiene esta carpeta: `cafe-aurora/`

Dentro habrá archivos simples:

`index.html`

`menu.txt`

`contacto.txt`

`README.md`

PARTE 1 Crear repositorio

1) git init

Inicializa Git.

```
git init
```

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/Cafe-Aurora ((ff828d3...))
$ git init
Initialized empty Git repository in C:/Users/Aprendiz/Desktop/Cafe-Aurora/.git/
```

Resultado:

```
Initialized empty Git repository
```



2) git status

Ver estado actual.

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/Cafe-Aurora (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
      README.md
      index.html
      menu.txt
```

git status

3) git config

Configurar usuario.

```
git config --global user.name "Juan"
git config --global user.email "juan@email.com"
```

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/Cafe-Aurora ((ff828d3...))
$ git config --global user.name "castillo"

Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/Cafe-Aurora ((ff828d3...))
$ git config --global user.email "juanse.castillo2008@gmail.com"
```

Ver configuración:

git config --list



```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/Cafe-Aurora (master)
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=castillo
user.email=juanse.castillo2008@gmail.com
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true
```

PARTE 2 Primer commit

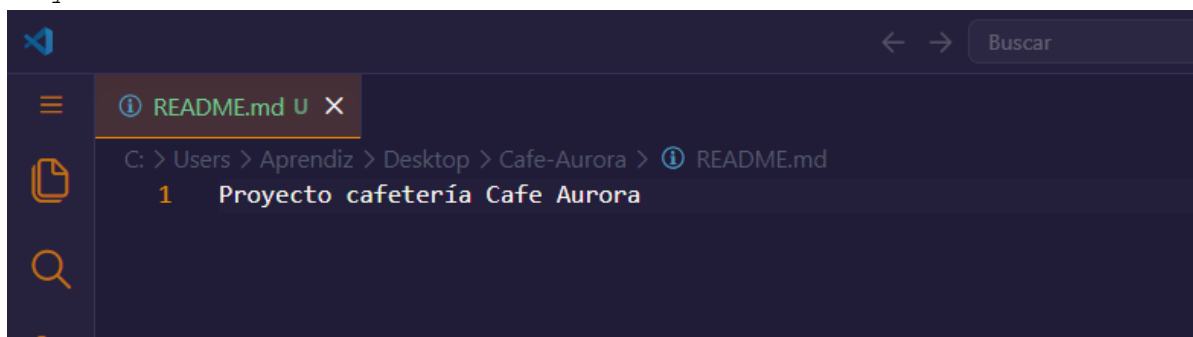
Crea:

[README.md](#)

 README.md	26/05/2026 8:58 p. m.	Archivo de origen ...	0 KB
---	-----------------------	-----------------------	------

Contenido:

Proyecto cafetería Cafe Aurora



4) git add

git add [README.md](#)



```
Apréndiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/Cafe-Aurora (master)
$ git add README.md
```

5) git status

git status

Verás:

Changes to be committed

```
Apréndiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/Cafe-Aurora (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   README.md

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        index.html
        menu.txt
```

6) git commit

git commit -m "Primer commit"

```
Apréndiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/Cafe-Aurora (master)
$ git commit -m "mi primer commit"
[master (root-commit) 4905024] mi primer commit
1 file changed, 1 insertion(+)
 create mode 100644 README.md
```

7) git log

git log

```
Apréndiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/Cafe-Aurora (master)
$ git log
commit 49050241b82f97a1b02e87238a2f992e25625ed1 (HEAD -> master)
Author: castillo <juanse.castillo2008@gmail.com>
Date:   Tue May 26 21:08:27 2026 -0500

    mi primer commit
```

8) git log --oneline



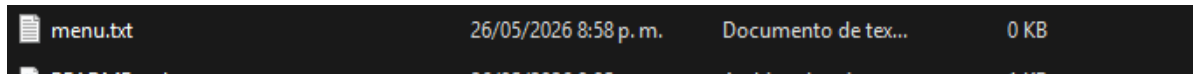
```
git log --oneline
```

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/Cafe-Aurora (master)
$ git log --oneline
4905024 (HEAD -> master) mi primer commit
```

PARTE 3 Trabajar con archivos

Crea:

menu.txt



9) git add .

```
git add .
```

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/Cafe-Aurora (master)
$ git add menu.txt
```

10) git diff

Modifica menu.txt.

Luego:

```
git diff
```

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/Cafe-Aurora (master)
$ git diff
diff --git a/menu.txt b/menu.txt
index e69de29..3f5a1b1 100644
--- a/menu.txt
+++ b/menu.txt
@@ -0,0 +1 @@
+primer proyecto :)
\ No newline at end of file
```

Muestra cambios no staged.

11) git commit

```
git commit -m "Agrega menú inicial"
```



```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/Cafe-Aurora (master)
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/Cafe-Aurora (master)
$ git commit -m "menu inicial"
[master 1861cae] menu inicial
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 menu.txt
```

PARTE 4 Branches

12) git branch

git branch

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/Cafe-Aurora (master)
$ git branch
* master
```

13) git branch feature/contacto

git branch feature/contacto

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/Cafe-Aurora (master)
$ git branch feature/contacto
```

14) git checkout

git checkout feature/contacto

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/Cafe-Aurora (master)
$ git branch feature/contacto

Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/Cafe-Aurora (master)
$ git checkout feature/contacto
M      menu.txt
Switched to branch 'feature/contacto'

Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/Cafe-Aurora (feature/contacto)
$
```

15) git switch

Volver a main:

git switch main

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/Cafe-Aurora (feature/contacto)
$ git switch master
M      menu.txt
Switched to branch 'master'
```



16) git checkout -b

Crear y cambiar:

```
git checkout -b feature/redes-sociales
```

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/Cafe-Aurora (master)
$ git checkout -b feature/redes-sociales
Switched to a new branch 'feature/redes-sociales'
```

PARTE 5 Merge

Agrega:

```
instagram: @cafeaurora
```

Commit:

```
git add .
git commit -m "Agrega redes sociales"
```

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/Cafe-Aurora (feature/redes-sociales)
$ git add .

Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/Cafe-Aurora (feature/redes-sociales)
$ git commit -m "instagram: @cafeaurora"
[feature/redes-sociales e405cb0] instagram: @cafeaurora
 2 files changed, 1 insertion(+)
 create mode 100644 index.html
```

17) git merge

Volver a main:

```
git switch main
```

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/Cafe-Aurora (feature/redes-sociales)
$ git switch master
Switched to branch 'master'

Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/Cafe-Aurora (master)
$
```

Fusionar:

```
git merge feature/redes-sociales
```



```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/Cafe-Aurora (master)
$ git merge feature/redes-sociales
Updating 1861cae..e405cb0
Fast-forward
 index.html | 0
 menu.txt   | 1 +
 2 files changed, 1 insertion(+)
 create mode 100644 index.html
```

PARTE 6 Eliminar y mover archivos

18) git rm

```
git rm contacto.txt
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/Desktop/cafe-aurora (master)
$ git rm contacto.txt
rm 'contacto.txt'
```

.git	27/5/2026 20:52	Carpeta de archivos	
index	27/5/2026 20:52	Microsoft Edge H...	0 KB
menu	27/5/2026 20:52	Documento de tex...	1 KB
README	27/5/2026 20:44	Archivo de origen ...	1 KB

19) git mv

```
git mv menu.txt menu-cafeteria.txt
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/Desktop/cafe-aurora (master)
$ git mv menu.txt menu-cafeteria.txt

Juan Castillo@DESKTOP-7050GLT MINGW64 ~/Desktop/cafe-aurora (master)
$
```

menu-cafeteria	27/5/2026 20:52	Documento de tex...	1 KB
----------------	-----------------	---------------------	------

20) git commit

```
git commit -m "Reorganiza archivos"
```

PARTE 7 Restaurar cambios

Modifica README.



```
1 Proyecto cafeteria cafe Aurora
2
3 modificacion readme
```

21) git restore

git restore [README.md](#)

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/Desktop/cafe-aurora (master)
$ git restore README.md
```

Descarta cambios.

```
1 Proyecto cafeteria cafe Aurora
```

22) git restore --staged

git restore --staged README.md

Quita del staging.

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/Desktop/cafe-aurora (master)
$ git restore --stage README.md
```

PARTE 8 Trabajar con remoto (GitHub)

23) git remote add

git remote add origin <https://github.com/juan/cafe-aurora.git>

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/Desktop/cafe-aurora (master)
$ git remote add origin https://github.com/juansecastillo2008-bot/Cafeteria-Aurora.git
```



24) git remote -v

```
git remote -v
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/Desktop/cafe-aurora (master)
$ git remote -v
origin  https://github.com/juansecastillo2008-bot/Cafeteria-Aurora.git (fetch)
origin  https://github.com/juansecastillo2008-bot/Cafeteria-Aurora.git (push)
```

25) git push

```
git push -u origin main
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/Desktop/cafe-aurora (master)
$ git push -u origin master
info: please complete authentication in your browser...
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 16 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (11/11), 1013 bytes | 1013.00 KiB/s, done.
Total 11 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/juansecastillo2008-bot/Cafeteria-Aurora.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
```

26) git pull

```
git pull
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/Desktop/cafe-aurora (master)
$ git pull
Already up to date.
```

27) git fetch

```
git fetch
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/Desktop/cafe-aurora (master)
$ git fetch
```

28) git clone

Otro desarrollador haría:

```
git clone https://github.com/juan/cafe-aurora.git
```



```
$ git clone https://github.com/juansecastillo2008-bot/Cafeteria-Aurora.git
Cloning into 'Cafeteria-Aurora'...
remote: Enumerating objects: 11, done.
remote: Counting objects: 100% (11/11), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 11 (delta 1), reused 11 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (11/11), done.
Resolving deltas: 100% (1/1), done.
```

PARTE 9 Stash

Estás trabajando y necesitas cambiar de rama rápido.

29) git stash

git stash

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/Desktop/cafe-aurora (master)
$ git stash
No local changes to save
```

Guarda temporalmente cambios.

30) git stash list

git stash list

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/Desktop/cafe-aurora (master)
$ git stash list
```

31) git stash pop

git stash pop

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/Desktop/cafe-aurora (master)
$ git stash pop
No stash entries found.
```

Recupera cambios.

PARTE 10 Historial y detalles

32) git show

git show



```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/Desktop/cafe-aurora (master)
$ git show
commit 62e273c1c0dab70ad074946e23fb49954a0eca85 (HEAD -> master, origin/master, origin/HEAD)
Author: castillo <juanse,castillo2008@gmail.com>
Date: Wed May 27 20:55:46 2026 +0200

    Reorganiza archivos

diff --git a/contacto.txt b/contacto.txt
deleted file mode 100644
index e69de29..0000000
diff --git a/menu.txt b/menu-cafeteria.txt
similarity index 100%
rename from menu.txt
rename to menu-cafeteria.txt
```

33) git reflog

git reflog

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/Desktop/cafe-aurora (master)
$ git reflog
62e273c (HEAD -> master, origin/master, origin/HEAD) HEAD@{0}: commit: Reorganiza archivos
b75e571 (feature/redes-sociales) HEAD@{1}: merge feature/redes-sociales: Fast-forward
27308de (feature/contacto) HEAD@{2}: checkout: moving from feature/redes-sociales to master
b75e571 (feature/redes-sociales) HEAD@{3}: commit: instagram: @cafeaurora
27308de (feature/contacto) HEAD@{4}: checkout: moving from master to feature/redes-sociales
27308de (feature/contacto) HEAD@{5}: checkout: moving from feature/contacto to master
27308de (feature/contacto) HEAD@{6}: checkout: moving from master to feature/contacto
27308de (feature/contacto) HEAD@{7}: commit: Agrega menu inicial
da57b04 HEAD@{8}: commit (initial): primer commit
```

Historial interno.

34) git blame

git blame [README.md](#)

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/Desktop/cafe-aurora (master)
$ git blame README.md
Ada57b04 (castillo 2026-05-27 20:45:37 +0200 1) Proyecto cafeteria cafe Aurora
```

Quién modificó cada línea.

PARTE 11 Tags y versiones

35) git tag

git tag v1.0

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/Desktop/cafe-aurora (master)
$ git tag v1.0
```



36) git tag -a

```
git tag -a v1.1 -m "Versión estable"
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/Desktop/cafe-aurora (master)
$ git tag -a v1.1 -m "version estable"
```

37) git push --tags

```
git push --tags
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/Desktop/cafe-aurora (master)
$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 166 bytes | 166.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/juansecastillo2008-bot/Cafeteria-Aurora.git
 * [new tag]          v1.0 -> v1.0
 * [new tag]          v1.1 -> v1.1
```

PARTE 12 Reset y revert

38) git reset

```
git reset HEAD README.md
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/Desktop/cafe-aurora (master)
$ git reset HEAD README.md
```

Quitar del staging.

39) git revert

```
git revert HEAD
```

Deshacer commit sin borrar historial.



```
Revert "Reorganiza archivos"

This reverts commit 62e273c1c0dab70ad074946e23fb49954a0eca85.

# Please enter the commit message for your changes. Lines starting
# with '#' will be ignored, and an empty message aborts the commit.
#
# On branch master
# Your branch is up to date with 'origin/master'.
#
# Changes to be committed:
#   new file:   contacto.txt
#   renamed:   menu-cafeteria.txt -> menu.txt
#
# Untracked files:
#   Cafeteria-Aurora/
#
~
~
~
~
```

PARTE 13 Rebase

40) git rebase

```
git rebase main
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/Desktop/cafe-aurora (master)
$ git rebase master
Current branch master is up to date.
```

Reorganiza historial.

EXTRA Flujo profesional

```
git pull
git switch -c feature/menu
git add .
git commit -m "Actualiza menú"
git push origin feature/menu
```

RESUMEN VISUAL

```
Working Directory
  ↓
git add
  ↓
Staging Area
  ↓
git commit
```



↓
Repositorio local
↓
git push
↓
GitHub

LOS 40 COMANDOS APRENDIDOS

- 1 git init
- 2 git status
- 3 git config
- 4 git add
- 5 git commit
- 6 git log
- 7 git log --oneline
- 8 git diff
- 9 git branch
- 10 git checkout
- 11 git switch
- 12 git merge
- 13 git rm
- 14 git mv
- 15 git restore
- 16 git remote
- 17 git push
- 18 git pull
- 19 git fetch
- 20 git clone
- 21 git stash
- 22 git stash list
- 23 git stash pop
- 24 git show
- 25 git reflog
- 26 git blame
- 27 git tag
- 28 git tag -a
- 29 git push --tags
- 30 git reset
- 31 git revert
- 32 git rebase
- 33 git checkout -b
- 34 git remote add



```
35 git restore --staged
36 git branch feature/*
37 git switch main
38 git merge feature/*
39 git add .
40 git commit -m
```

¿Te atreves a realizar el siguiente reto?

Una **simulación real de empresa**, donde usaras:

- comandos básicos (add, commit, branch...)
- comandos intermedios (stash, rebase, reset...)
- comandos avanzados (reflog, bisect, cherry-pick...)
- inspección interna de Git
- herramientas de debugging
- configuración
- limpieza
- flujos de equipo

E-commerce Café Aurora pro

Trabaja en equipo con GitHub. Para desarrollar:

- tienda online
- pagos
- menú
- sistema de usuarios
- corrección de bugs

PARTE 1 CONFIGURACIÓN INICIAL (10 comandos)

```
git --version
git config --global user.name "Juan"
git config --global user.email "juan@email.com"
git config --list
git help
git help commit
git init
git clone https://github.com/cafe/aurora.git
git remote -v
git remote add origin URL
```




PARTE 2 INSPECCIÓN DEL PROYECTO (10 comandos)

```
git status
git status -s
git log
git log --oneline
git log --graph
git log --all
git diff
git diff HEAD
git diff --staged
git show HEAD
```

PARTE 3 WORKFLOW BÁSICO (10 comandos)

```
git add archivo.txt
git add .
git add -A
git commit -m "mensaje"
git commit -am "mensaje"
git rm archivo.txt
git mv viejo.txt nuevo.txt
git restore archivo.txt
git restore --staged archivo.txt
git clean -f
```

PARTE 4 RAMAS (12 comandos)

```
git branch
git branch feature/login
git branch -d feature/login
git branch -D feature/login
git checkout main
git checkout -b feature/login
git switch main
git switch -c feature/login
git merge feature/login
git merge --no-ff feature/login
git branch -m nuevo-nombre
git branch -r
```

PARTE 5 REBASE Y HISTORIAL (8 comandos)

```
git rebase main
git rebase -i HEAD~3
git rebase --abort
git rebase --continue
git cherry-pick abc1234
git revert HEAD
```



```
git reset HEAD~1  
git reset --soft HEAD~1
```

PARTE 6 REMOTO (10 comandos)

```
git push  
git push -u origin main  
git push origin feature/login  
git pull  
git fetch  
git fetch --all  
git remote add origin URL  
git remote remove origin  
git remote show origin  
git ls-remote
```

PARTE 7 STASH (6 comandos)

```
git stash  
git stash list  
git stash pop  
git stash apply  
git stash drop  
git stash clear
```

PARTE 8 RECUPERACIÓN Y DEBUG (10 comandos)

```
git reflog  
git reset --hard HEAD  
git reset --mixed HEAD  
git reset --soft HEAD  
git fsck  
git gc  
git blame archivo.txt  
git show commit-id  
git bisect start  
git bisect bad  
git bisect Good
```

PARTE 9 VERSIONES (5 comandos)

```
git tag v1.0  
git tag -a v1.1 -m "release"  
git tag  
git push --tags  
git show v1.0
```

PARTE 10 LIMPIEZA (5 comandos)

```
git clean -n  
git clean -f  
git clean -fd  
git prune  
git gc -aggressive
```



PARTE 11 UTILIDADES AVANZADAS (5 comandos)

```
git shortlog
git archive
git describe
git check-ignore
git update-index
```

PARTE 12 AYUDA Y CONTROL (5 comandos)

```
git help
git help merge
git help rebase
git help stash
git config --global -l
```

PARTE 13 FLUJO REAL DE TRABAJO (EQUIPO)

Ejemplo completo de trabajo profesional:

```
git pull
git checkout -b feature/payments
git add .
git commit -m "Add payments system"
git push origin feature/payments
git fetch
git rebase main
git merge feature/payments
git push
```

RESUMEN

En empresas se usa realmente esto:

20% de comandos hacen 80% del trabajo:

- git add
- git commit
- git push
- git pull
- git branch
- git checkout / switch
- git merge



- git stash
- git log
- git reset

IMPORTANTE

Muchos comandos:

- se usan solo en casos extremos (bisect, fsck, gc)
- o en debugging
- o en equipos grandes

CONCLUSIÓN

Git no se aprende por cantidad de comandos, sino por entender flujo:

Working Directory → Staging → Commit → Remote

Taller Práctico de Git

Proyecto: Café Aurora

“Aprender Git no es memorizar comandos, es entender cómo trabajan los equipos reales de desarrollo.”

Objetivo del Taller

Construiremos paso a paso el proyecto Café Aurora, una página web sencilla para una cafetería ficticia.

Durante el proceso aprenderás cómo se trabaja con Git en ambientes profesionales usando:

- Commits
- Ramas (branches)
- Merge
- Stash
- Rebase
- Tags
- GitHub y remotos
- Restauración de archivos



- Historial y debugging
- Resolución de conflictos
- Flujo de trabajo colaborativo

Al finalizar el taller ¿qué aprenderás realmente?

- Crear repositorios profesionales
- Controlar versiones de proyectos
- Trabajar en equipo usando GitHub
- Recuperar cambios perdidos
- Resolver errores comunes
- Entender el flujo real de desarrollo en empresas

EXTRA Flujo profesional real

```
git pull
git switch -c feature/menu
git add .
git commit -m "Actualiza menú"
git push origin feature/menu
```

RESUMEN VISUAL DEL FLUJO GIT

```
Working Directory
  ↓
git add
  ↓
Staging Area
  ↓
git commit
  ↓
Repositorio local
  ↓
git push
  ↓
GitHub
```

LOS COMANDOS MÁS IMPORTANTES

El 80% del trabajo real usa:

```
git add
git commit
git push
git pull
```



```
git branch
git switch
git merge
git stash
git log
git reset
```

IMPORTANTE

Muchos comandos avanzados como `git bisect`, `git fsck`, `git gc`
Se usan principalmente para:

- debugging
- mantenimiento
- proyectos grandes
- recuperación avanzada

CONCLUSIÓN

Git no se aprende memorizando 100 comandos.

Git se aprende entendiendo el flujo:

```
Working Directory
  ↓
Staging
  ↓
Commit
  ↓
Remote Repository
```

RETO FINAL

Proyecto: E-commerce Café Aurora Pro

Trabajarás como si estuvieras en una empresa real usando:

- GitHub
- ramas por funcionalidades
- corrección de bugs
- trabajo en equipo
- rebase y merge
- debugging
- recuperación de errores



- versiones estables

Desafío Final

¿Serías capaz de:

- crear nuevas funcionalidades?
- resolver conflictos?
- recuperar commits eliminados?
- trabajar en equipo sin dañar el proyecto?
- desplegar versiones estables?

Meta del Taller

Al terminar este laboratorio podrás decir:

- “Sé usar Git en proyectos reales.”
- “Entiendo cómo trabajan los equipos de desarrollo.”
- “Puedo colaborar profesionalmente usando Git y GitHub.”

Objetivo del proyecto

Construir una tienda online para:

- vender café
- mostrar catálogo
- gestionar pedidos
- manejar usuarios
- integrar pagos
- publicar releases

ARQUITECTURA DEL REPOSITORIO

```
cafe-aurora/  
├── frontend/  
├── backend/  
├── docs/  
├── tests/  
├── .gitignore  
└── README.md
```



.git	29/5/2026 15:27	Carpeta de archivos	
backend	29/5/2026 07:28	Carpeta de archivos	
docs	29/5/2026 07:29	Carpeta de archivos	
frontend	29/5/2026 08:11	Carpeta de archivos	
test	29/5/2026 07:30	Carpeta de archivos	
gitignore	29/5/2026 07:30	Archivo	0 KB
README	29/5/2026 07:30	Archivo de origen ...	0 KB

FASE 1 INICIO DEL PROYECTO

1. Clonar o iniciar repo

```
git clone https://github.com/cafe/aurora.git
cd cafe-aurora
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
$ git clone https://github.com/juansecastillo2008-bot/cafe-aurora-definitivo.git
Cloning into 'cafe-aurora-definitivo'...
remote: Enumerating objects: 13, done.
remote: Counting objects: 100% (13/13), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 13 (delta 4), reused 13 (delta 4), pack-reused 0 (from 0)
Receiving objects: 100% (13/13), done.
Resolving deltas: 100% (4/4), done.
```

o si empiezas desde cero:

```
git init
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo
$ git init
Initialized empty Git repository in C:/Users/Juan Castillo/cafe-aurora-definitivo/.git/
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
```

2. Configuración del equipo

```
git config --global user.name "Developer 1"
git config --global user.email "dev1@cafe.com"
```




```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/cafe-aurora-reto (master)
$ git config --global user.name "juan castillo"

Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/cafe-aurora-reto (master)
$ git config --global user.email "juanse.castillo2008@gmail.com"

Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/cafe-aurora-reto (master)
$ |
```

3. Conectar remoto

```
git remote add origin https://github.com/cafe/aurora.git
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
$ git remote add origin https://github.com/juansecastillo2008-bot/cafe-aurora-definitivo.git
```

FASE 2 PRIMER RELEASE BASE

4. Crear estructura inicial

```
git status
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   README.md
    new file:   backend/pagos.js
    new file:   backend/server.txt
    new file:   frontend/catalogo.txt
    new file:   frontend/checkout.html
    new file:   frontend/index.html
    new file:   gitignore
```

5. Agregar todo

```
git add .
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
$ git add .
```

6. Primer commit

```
git commit -m "chore: estructura inicial del proyecto"
```



```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
$ git commit -m "chore: estructura inicial del proyecto"
[master (root-commit) ebe5a1d] chore: estructura inicial del proyecto
7 files changed, 0 insertions(+), 0 deletions(-)
create mode 100644 README.md
create mode 100644 backend/pagos.js
create mode 100644 backend/server.txt
create mode 100644 frontend/catalogo.txt
create mode 100644 frontend/checkout.html
create mode 100644 frontend/index.html
create mode 100644 gitignore
```

7. Subir a GitHub

```
git push -u origin main
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
$ git push origin master
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 16 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (13/13), 1015 bytes | 507.00 KiB/s, done.
Total 13 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), done.
To https://github.com/juansecastillo2008-bot/cafe-aurora-definitivo.git
 * [new branch]      master -> master
```

FASE 3 DESARROLLO POR RAMAS

8. Crear rama de desarrollo

```
git checkout -b develop
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
$ git checkout -b develop
Switched to a new branch 'develop'

Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (develop)
$ git checkout -b feature/catalogo
```

9. Crear feature catálogo

```
git checkout -b feature/catalogo
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (develop)
$ git checkout -b feature/catalogo
Switched to a new branch 'feature/catalogo'

Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (feature/catalogo)
$ echo "cafe colombiano\cafe espresso\cafe premium" > catalogo.txt
```

10. Trabajar catálogo

Se agregan productos:



- café colombiano
- café espresso
- café premium

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (feature/catalogo)
$ echo "cafe clombiano\cafe espresso\cafe premium" catalogo.txt
cafe clombiano\cafe espresso\cafe premium catalogo.txt

Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (feature/catalogo)
$ echo "cafe clombiano" >>frontend/catalogo.txt

Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (feature/catalogo)
$ echo "cafe espresso" >>frotend/catalogo.txt
bash: frotend/catalogo.txt: No such file or directory

Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (feature/catalogo)
$ echo "cafe espresso" >>frontend/catalogo.txt

Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (feature/catalogo)
$ echo "cafe premium" >>frotend/catalogo.txt
bash: frotend/catalogo.txt: No such file or directory

Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (feature/catalogo)
$ echo "cafe premium" >>frontend/catalogo.txt
```

11. Guardar cambios

```
git add .
git commit -m "feat: catálogo de productos inicial"
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
$ git add .

Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
$ git commit -m "catalogo productos inicial"
[master c7feb90] catalogo productos inicial
1 file changed, 3 insertions(+)
```

12. Subir rama

```
git push origin feature/catalogo
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
$ git push origin feature/catalogo
Everything up-to-date
```

13. Merge a develop

```
git checkout develop
git merge feature/catalogo
```



```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
$ git checkout develop
Switched to branch 'develop'

Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (develop)
$ git merge feature/catalogo
Already up to date.
```

FASE 4 SISTEMA DE PAGOS

14. Crear rama

```
git checkout -b feature/pagos
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (develop)
$ git checkout -b feature/pagos
Switched to a new branch 'feature/pagos'
```

15. Cambios en pagos

Se agregan:

- PayPal
- tarjeta
- Stripe

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (feature/pagos)
$ echo "paypal">> frontend/checkout.html

Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (feature/pagos)
$ echo "tarjeta">> frontend/checkout.html

Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (feature/pagos)
$ echo "efecty">> frontend/checkout.html
```

16. Guardar cambios

```
git add .
git commit -m "feat: integración de pagos"
```



```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
$ git add .

Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
$ git commit -m "integracion de pagos"
[master 9f1bc80] integracion de pagos
1 file changed, 3 insertions(+)
```

17. Subir rama

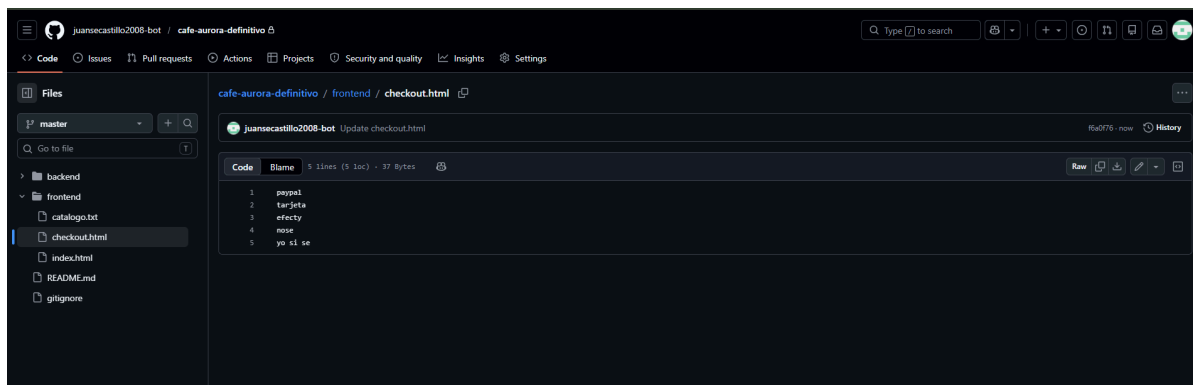
`git push origin feature/pagos`

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
$ git push origin feature/pagos
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/pagos' on GitHub by visiting:
remote:   https://github.com/juansecastillo2008-bot/cafe-aurora-definitivo/pull/new/feature/pagos
remote:
To https://github.com/juansecastillo2008-bot/cafe-aurora-definitivo.git
 * [new branch]      feature/pagos -> feature/pagos
```

FASE 5 CONFLICTO REAL (IMPORTANTE)

Dos personas modifican el mismo archivo:

`checkout.html`





Commits			
History for <code>cafe-aurora-definitivo / frontend / checkout.html</code> on <code>master</code>			
All users All time			
Commits on May 29, 2026			
Update checkout.html	Verified	d70a95e	<>
Update checkout.html	Verified	9b1e9f7	<>
edicion parte 1 tebanju 'nose 2'	Verified	d1a6511	<>
Update checkout.html	Verified	1f6a076	<>
Update checkout.html	Verified	1c77b81	<>
integracion de pagos		9f1bc80	<>
chore: estructura inicial del proyecto		ebe5ald	<>
End of commit history for this file			

18. Merge conflict

```
git checkout develop
git merge feature/pagos
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
$ git checkout develop
Switched to branch 'develop'

Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (develop)
$ git merge feature/pagos
Already up to date.
```

Conflicto aparece.

19. Ver estado

```
git status
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
$ git status
On branch master
nothing to commit, working tree clean
```

20. Resolver conflicto manualmente

Editas archivo y decides versión correcta.

21. Marcar como resuelto



```
git add checkout.html
git commit -m "fix: resolve merge conflict en pagos"
```

FASE 6 RELEASE

22. Crear rama release

```
git checkout -b release/v1.0
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (release/v1.0)
$
```

23. Ajustes finales

- bugs menores
- UI
- performance

A screenshot of a Windows terminal window titled "MINGW64: c:/Users/Juan Castillo/cafe-aurora-definitivo". The terminal shows the user making three updates to the codebase in the 'release/v1.0' branch. The commands and their outputs are as follows:

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (release/v1.0)
$ echo "Arreglo de bug menor en carga" >> backend/server.txt

Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (release/v1.0)
$ echo "Ajuste de diseño en la interfaz" >> frontend/index.html

Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (release/v1.0)
$ echo "optimizacion de velocidad de respuesta" >> backend/server.txt

Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (release/v1.0)
$ |
```

24. Commit final

```
git commit -am "release: versión 1.0 lista"
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (release/v1.0)
$ git commit -am "release: versión 1.0 lista"
warning: in the working copy of 'backend/server.txt', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'frontend/index.html', LF will be replaced by CRLF the next time Git touches it
[release/v1.0 49a65a6] release: versión 1.0 lista
 2 files changed, 3 insertions(+)
```

25. Merge a main

```
git checkout main
git merge release/v1.0
```



26. Tag de versión

```
git tag -a v1.0 -m "Primera versión estable"
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
$ git tag -a v1.0 -m "primera version estable"
```

27. Subir todo

```
git push origin main
git push --tags
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
$ git push origin master --force
Enumerating objects: 13, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 16 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (9/9), 887 bytes | 887.00 KiB/s, done.
Total 9 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To https://github.com/juansecastillo2008-bot/cafe-aurora-definitivo.git
+ d70a95e...192a487 master -> master (forced update)
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 171 bytes | 171.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/juansecastillo2008-bot/cafe-aurora-definitivo.git
* [new tag]          v1.0 -> v1.0
```

FASE 7 TRABAJO DE EMERGENCIA (HOTFIX)

28. Bug crítico en pagos

```
git checkout -b hotfix/pagos-error
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
$ git checkout -b hotfix/pagos-error
Switched to a new branch 'hotfix/pagos-error'

Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (hotfix/pagos-error)
```

29. Fix rápido

```
git commit -am "hotfix: error crítico en checkout"
```




30. Merge inmediato

```
git checkout main
```

```
$ git checkout master
Switched to branch 'master'

Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
```

```
git merge hotfix/pagos-error
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
$ git merge hotfix/pagos-error
Already up to date.

Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
```

FASE 8 HISTORIAL Y CONTROL

31. Ver historial

```
git log --oneline --graph
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
$ git log --oneline --graph
* 192a487 (HEAD -> master, origin/master, hotfix/pagos-error) fix: incluir cambios de pago en develop
* 9f1bc80 (tag: v1.0) integracion de pagos
* c7feb90 catalogo productos inicial
* eb5a1d (origin/feature/pagos, origin/feature/catalogo, feature/pagos, feature/catalogo, develop) chore: estructura inicial del proyecto

Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
```

32. Ver cambios de un commit

```
git show HEAD
```

```
$ git show HEAD
commit 192a48794d1a365bdef13b15834ad9491ed82bee (HEAD -> master, origin/master, hotfix/pagos-error)
Author: castillo <juanse,castillo2008@gmail.com>
Date: Fri May 29 17:13:08 2026 +0200

    fix: incluir cambios de pago en develop

diff --git a/frontend/checkout.html b/frontend/checkout.html
index 0a83eb5..e69de29 100644
--- a/frontend/checkout.html
+++ b/frontend/checkout.html
@@ -1,3 +0,0 @@
-paypal
-tarjeta
-efecty
```

33. Quién modificó qué

```
git blame backend/payments.js
```

34. Recuperar estado anterior



git reflog

FASE 9 STASH (trabajo urgente)

35. Guardar cambios temporales

git stash

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
$ git stash
No local changes to save
```

36. Volver cambios

git stash pop

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
$ git stash pop
No stash entries found.
```

FASE 10 REBASE (historial limpio)

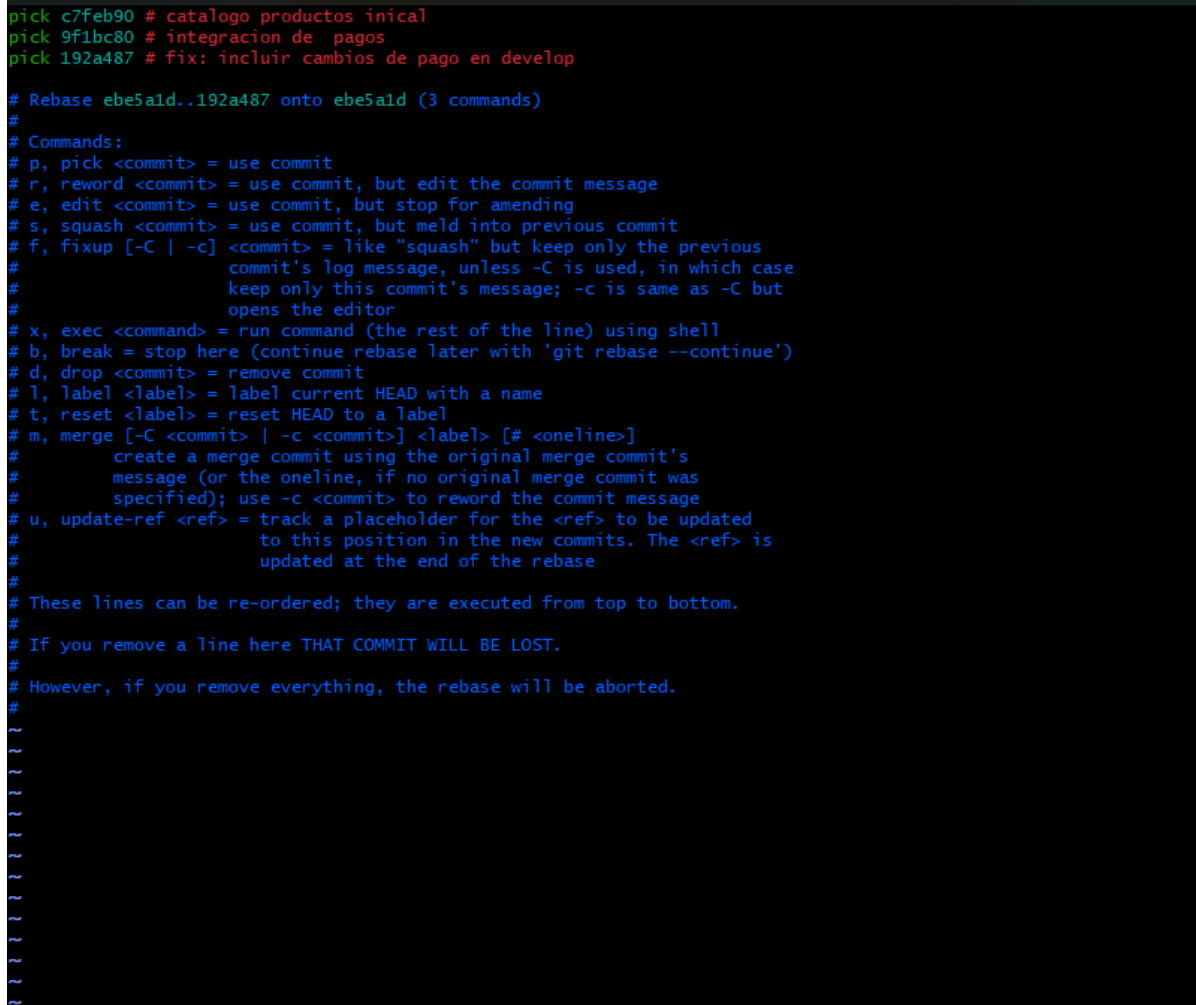
37. Reorganizar commits

git rebase main

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master)
$ git rebase master
Current branch master is up to date.
```

38. Rebase interactivo

git rebase -i HEAD~3



39. Buscar bug

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master|REBASE)
$ git bisect start
status: waiting for both good and bad commits
```

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master|REBASE)
$ git bisect bad
status: waiting for good commit(s), bad commit known
```

Ing. Víctor Rincón



40. Limpiar repo

git gc

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master|REBASE)
$ git gc
Enumerating objects: 21, done.
Counting objects: 100% (21/21), done.
Delta compression using up to 16 threads
Compressing objects: 100% (17/17), done.
Writing objects: 100% (21/21), done.
Total 21 (delta 6), reused 0 (delta 0), pack-reused 0 (from 0)
```

git fsck

```
Juan Castillo@DESKTOP-7050GLT MINGW64 ~/cafe-aurora-definitivo (master|REBASE)
$ git fsck
Checking ref database: 100% (1/1), done.
Checking object directories: 100% (256/256), done.
Checking objects: 100% (21/21), done.
Verifying commits in commit graph: 100% (5/5), done.
```

FLUJO FINAL DEL PROYECTO

```
feature → develop → release → main → tag
    ↓           ↓           ↓           ↓
  testing   staging   producción   versión
```

LO QUE APRENDISTE REALIZANDO EL ANTERIOR PROYECTO

Flujo real de empresa

- feature branches
- develop branch
- release branch
- hotfix branch

Trabajo en equipo

- merges
- conflictos
- rebase



Producción real

- tags
- versiones
- hotfix

Control total

- logs
- reflog
- blame
- bisect